

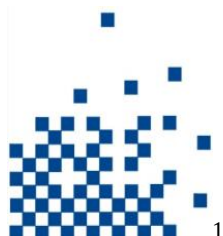


NAP874G
硬件测试报告

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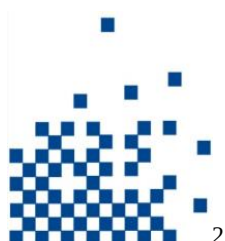
审核：王朋刚

核准：刘松辉

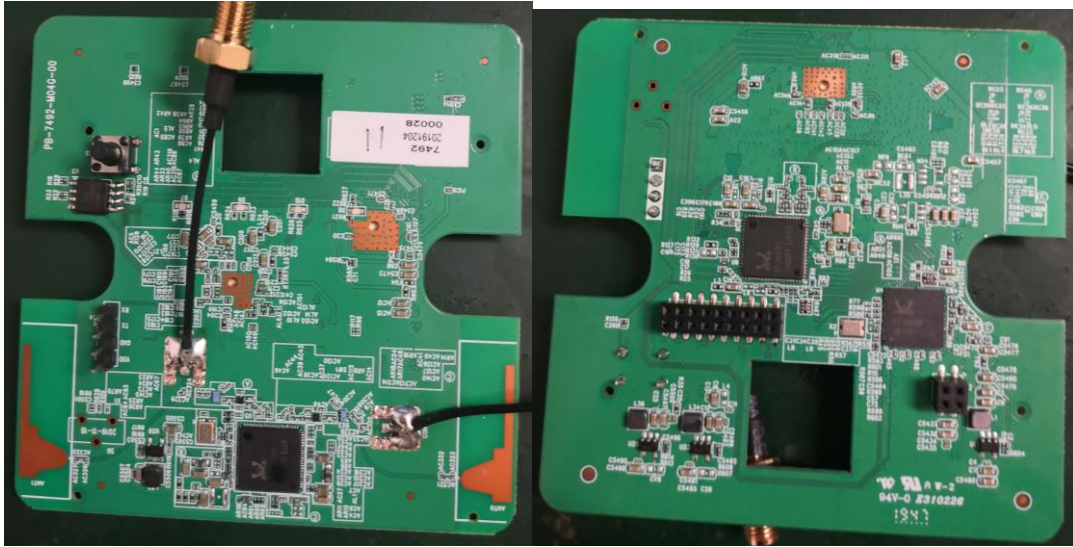


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一、测试基本信息

PCB 版本编号	PB-7492-M04G-00
机构编号	7492
芯片方案	RTL8812BRH+RTL8197FS-VE5-CG+RTL8363NB-VB-CG
测试时间	2020-4-23
测试样机规格	① 2.4GHZ/5GHZ 双频无线 AC 千兆路由器 ② 2.4G 最高速率 300Mbps,5G 最高速率 867Mbps。
测试样机软件版本	烧录文件: CMCC-R2ps-2-W.00-M-v1.0.0-190603_20190603.img TFTP 升级软件: fw_97f_8812brh_8363nb.bin
测试样机板数量	23PCS
测试地点	开发一部实验室
测试设备	IQXEL,示波器, 万用表, 屏蔽箱, 高温箱
附加说明	管理地址: http://192.168.8.1 产测程序: RTL8192E、RTL8812A
产品图片	

二、测试项目及结果

测试项目			是否测试	测试结果
电源	AC-DC	输入输出电压	/	/



		输入电压范围	/	/
		输入输出电流	/	/
		输入输出纹波	/	/
		输入输出功率	/	/
	DC-DC	输入输出电压	✓	✓
		输入电压范围	✓	✓
		输入输出电流	✓	✓
		输入输出纹波	✓	✓
		输入输出功率	✓	✓
信号测量	上电顺序	顺序	✓	✓
	时钟	频率	✓	✓
		幅度	✓	✓
		抖动	✓	✓
	RF	CCK RX (灵敏度)	✓	✓
		CCK TX (POW、EVM、频偏)	✓	✓
		OFDM RX (灵敏度)	✓	✓
		OFDM TX (POW、EVM、频偏)	✓	✓
		MCS20 TX (灵敏度)	✓	✓
		MCS20 TX (POW、EVM、频偏)	✓	✓
		MCS20 TX (灵敏度)	✓	✓
		MCS20 TX (POW、EVM、频偏)	✓	✓
		Beacon 功率	✓	✓
	数据信号	各接口数据	✓	✓
基本功能	串口	端口速率, 可操作性	✓	✓
	按键	作用是否有效	✓	✓
	指示灯	亮的方式	✓	✓
性能	无线性能	无线吞吐量测试	✓	✓
		屏蔽箱下衰减吞吐量测试	✓	✓
		室内覆盖测试	✓	✓
		天线场型测试	✓	✓
	有线端口交换		✓	✓
环境测试	烧机		✓	✓
	冷启动测试		✓	✓
	DUT 高低温功能及温度		✓	✓
上下点测试	上下电测试		✓	✓



三三、测试记录

1、电源

1.1、电源规格

1) 电源规格：53V/1.35A

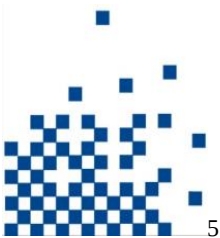
1.2、DC TO DC

测试点	标准	空载				满载				
	电压(V)	电压(V)	最大/平均电流(A)	纹波(mV)	平均功率(W)	电压(V)	最大/平均电流(A)	纹波(mV)	开关频率(HZ)	平均功率(W)
电源处	50.35~55.65	52.98	0.194/0.139	/	7.36	52.98	0.245/0.161	/	/	8.530
C26~C24	11.4~12.6	12.12	0.987/0.434	368	5.26	12.22	1.219/0.517	576	/	6.318
L5	3.135~3.465	3.307	1.846/1.290	52	4.266	3.314	2.421/1.842	76	1.02M	6.104
C25~C5497	1.71~1.89	1.807	0.312/0.142	48	0.257	1.805	0.329/0.173	60	1.658M	0.312
L4	0.95~1.05	1.049	0.633/0.401	52	0.421	1.043	0.777/0.527	66	1.561M	0.550

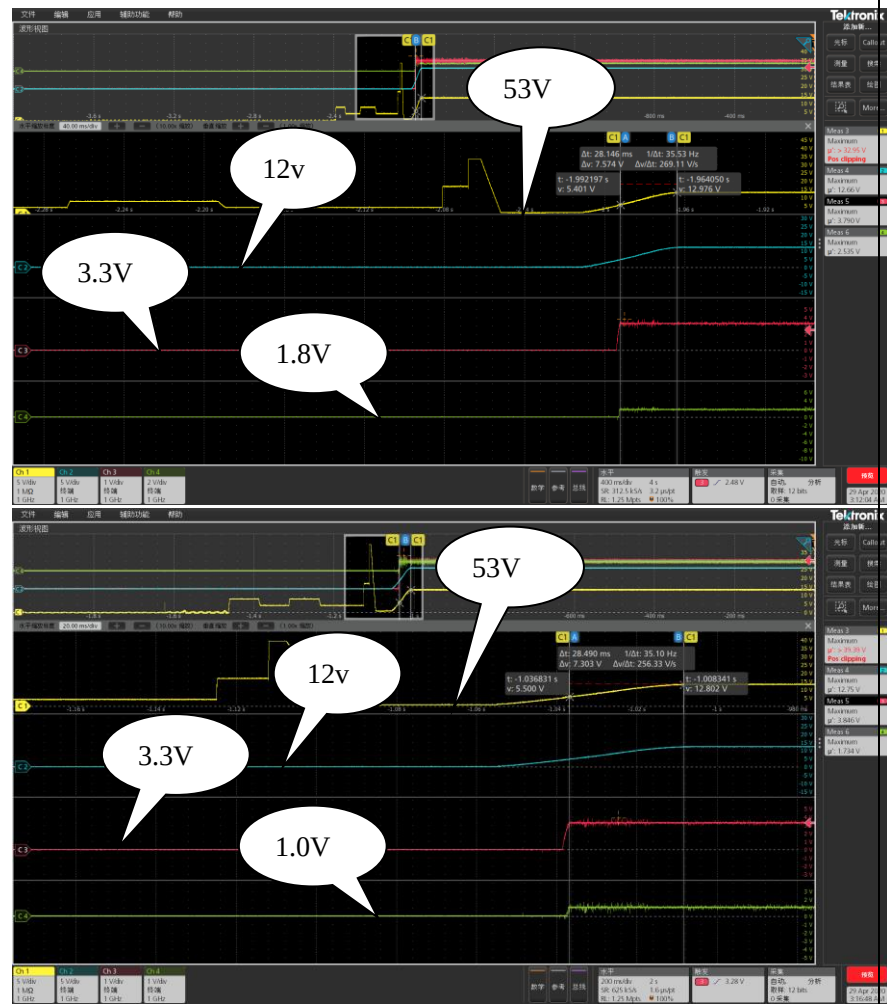
2、信号测量

2.1、数字信号

信号名称	频率	图片
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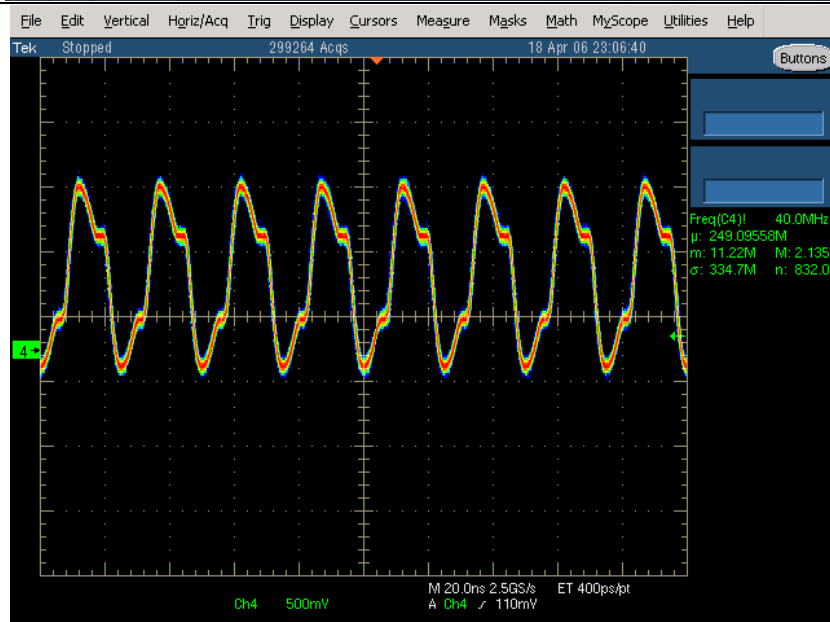


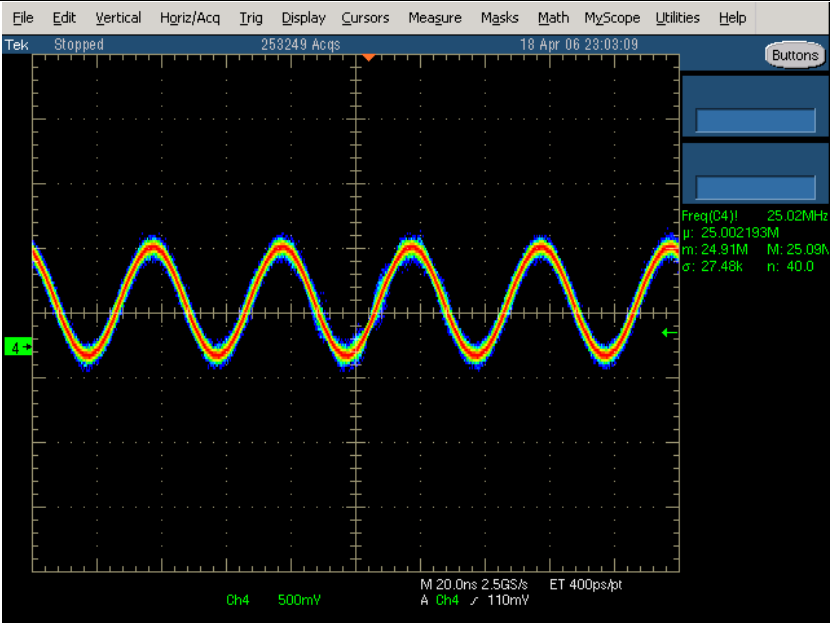
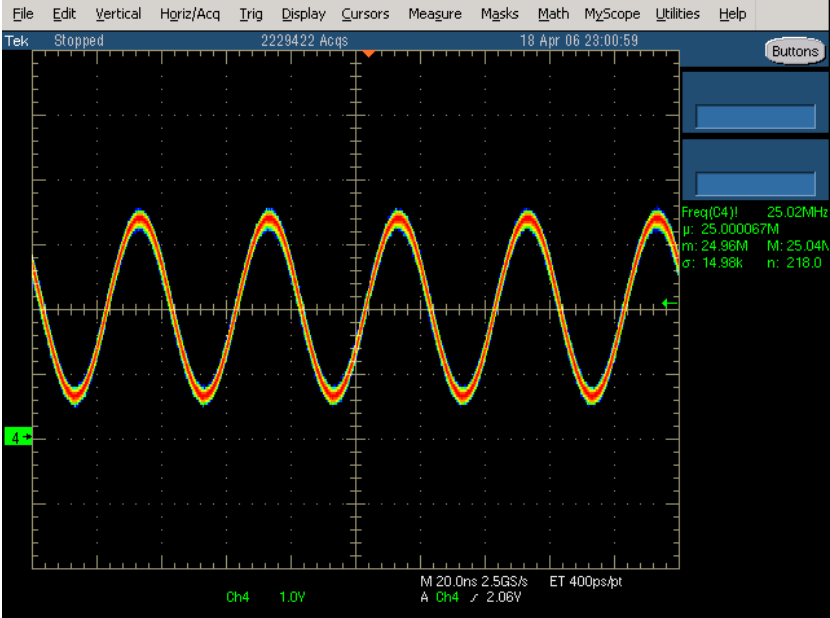
上电顺序



晶振
(U5)

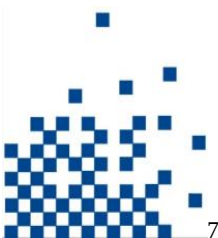
40MHZ



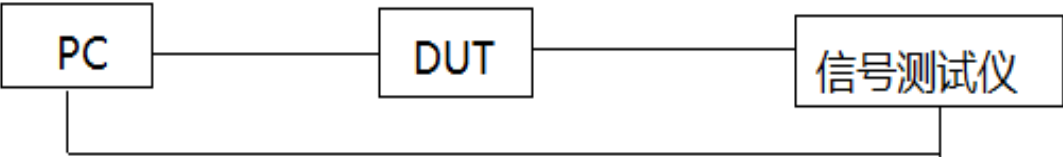
晶振 (U6)	25MHZ	
晶振 (U4)	24.99923 MHZ	

2.2、wifi 信号

2) RF 测量



拓扑结构



测试记录

RF 校准测试											
测试前提	测试数据必须是测试至少 5 片样机后 RF 一致性 OK 的数据										结论
2G TX	antenna	A		datarate	CCK-1	antenna	B		datarate	CCK-1	
	channel	1	7	13		channel	1	7	13		
	Power	22.23	22.14	22.22		Power	22.31	22.23	22.26		
	Evm	-28.12	-27.94	-27.41		Evm	-29.13	-28.92	-29.10		
	ppm	+10.87	+10.47	+10.43		ppm	+10.49	+10.50	+10.41		
	Mask	0	0.17	0		Mask	0.50	0.92	1.25		
	antenna	A		datarate	CCK-11	antenna	B		datarate	CCK-11	
	channel	1	7	13		channel	1	7	13		
	Power	22.08	22.04	22.0		Power	22.12	22.02	22.09		
	Evm	-23.66	-25.20	-24.12		Evm	-26.19	-25.30	-24.62		
	ppm	+10.83	+10.49	+10.37		ppm	+10.42	+10.37	+10.33		
	Mask	0	0	0		Mask	0	0	0.17		
	antenna	A		datarate	OFDM-54	antenna	B		datarate	OFDM-54	
	channel	1	7	13		channel	1	7	13		
	Power	16.2	16.18	16.29		Power	16.21	16.34	16.02		
	Evm	-26.80	-28.9	-30.03		Evm	-30.95	-31.11	-31.68		
	ppm	+11.15	+10.21	+10.98		ppm	+10.13	+10.18	+10.29		
	Mask	0	0	0		Mask	0	0	0		
	antenna	A		datarate	HT20-MCS0	antenna	B		datarate	HT20-MCS0	
	channel	1	7	13		channel	1	7	13		
	Power	14.90	14.96	14.56		Power	14.98	14.68	14.85		
	Evm	-28.97	-30.78	-32.66		Evm	-31.30	-31.66	-33.41		
	ppm	+10.63	+10.27	+10.70		ppm	+10.35	+10.53	+10.37		
	Mask	0	0	0		Mask	0	0	0		



antenna	A		datarate	HT20-MCS3	antenna	B		datarate	HT20-MCS3
channel	1	7	13		channel	1	7	13	
Power	15.21	15.18	14.80		Power	15.21	14.89	14.99	
Evm	-27.66	-30.01	-31.47		Evm	-31.67	-32.12	-32.84	
ppm	+11.01	+10.74	+10.64		ppm	+10.45	+10.62	+10.49	
Mask	0	0	0		Mask	0	0	0	
antenna	A		datarate	HT20-MCS7	antenna	B		datarate	HT20-MCS7
channel	1	7	13		channel	1	7	13	
Power	15.33	15.35	14.93		Power	15.28	14.93	15.09	
Evm	-28.02	-29.74	-31.75		Evm	-31.82	-32.28	-32.91	
ppm	+10.9	+10.9	+10.72		ppm	+10.41	+10.47	+10.38	
Mask	0	0	0		Mask	0	0	0	
antenna	A		datarate	HT40-MCS0	antenna	B		datarate	HT40-MCS0
channel	3	7	11		channel	3	7	11	
Power	14.71	14.73	14.56		Power	14.76	14.94	14.99	
Evm	-31.38	-31.97	-32.58		Evm	-32.70	-32.72	-32.79	
ppm	+10.54	+10.79	+10.63		ppm	+10.32	+10.27	+10.37	
Mask	0	0	0		Mask	0	0	0	
antenna	A		datarate	HT40-MCS3	antenna	B		datarate	HT40-MCS3
channel	3	7	11		channel	3	7	11	
Power	15.10	15.02	14.83		Power	14.99	15.23	15.27	
Evm	-29.62	-31.32	-32.19		Evm	-31.25	-31.65	-32.01	
ppm	+10.71	+10.68	+10.89		ppm	+10.32	+10.33	+10.47	
Mask	0	0	0		Mask	0	0	0	
antenna	A		datarate	HT40-MCS7	antenna	B		datarate	HT40-MCS7
channel	3	7	11		channel	3	7	11	
Power	15.34	15.30	15.09		Power	15.16	15.33	15.35	
Evm	-30.13	-30.00	-33.08		Evm	-31.87	-31.68	-32.24	
ppm	+10.19	+10.22	+10.5		ppm	+10.39	+10.21	+10.16	
Mask	0	0	0		Mask	0	0	0	



2G
RX

标准: PktNum:1000; PER(%): [0.00 ~ 8.00]											
Antenna:A Datarate: CCK-11						Antenna:B Datarate: CCK-11					
channel	1	7	13			channel	1	7	13		
SigLevel	-86	-86	-86			SigLevel	-86	-86	-86		
PER(%)	3.10	1.30	1.40			PER(%)	2.2	1.9	1.7		
标准: PktNum:1000; PER(%): [0.00 ~ 8.00]											
Antenna:A Datarate: CCK-1						Antenna:B Datarate: CCK-1					
channel	1	7	13			channel	1	7	13		
SigLevel	-94	-94	-94			SigLevel	-94	-94	-94		
PER(%)	3.3	2.2	2.6			PER(%)	1.8	0.9	1.0		
标准: PktNum:1000; PER(%): [0.00 ~ 10.00]											
Antenna:A Datarate: OFDM-54						Antenna:B Datarate: OFDM-54					
channel	1	7	13			channel	1	7	13		
SigLevel	-72	-72	-72			SigLevel	0.3	5	0.3		
PER(%)	0.1	0.0	0			PER(%)	0.3	5	0.3		
标准: PktNum:1000; PER(%): [0.00 ~ 10.00]											
Antenna:A Datarate: HT20-MCS0						Antenna:B Datarate: HT20-MCS0					
channel	1	7	13			channel	1	7	13		
SigLevel	-88	-88	-88			SigLevel	-88	-88	-88		
PER(%)	0.6	0.1	0.2			PER(%)	2.0	0.4	0		
标准: PktNum:1000; PER(%): [0.00 ~ 10.00]											
Antenna:A Datarate: HT20-MCS7						Antenna:B Datarate: HT20-MCS7					
channel	1	7	13			channel	1	7	13		
SigLevel	-70	-70	-70			SigLevel	-70	-70	-70		
PER(%)	0.3	0	0.2			PER(%)	0.5	0.5	2.2		
标准: PktNum:1000; PER(%): [0.00 ~ 10.00]											
Antenna:A Datarate: HT40-MCS0						Antenna:B Datarate: HT40-MCS0					
channel	1	7	13			channel	1	7	13		
SigLevel	-87	-87	-87			SigLevel	-87	-87	-87		
PER(%)	0.0	0.1	0			PER(%)	0.1	0.0	0.0		
标准: PktNum:1000; PER(%): [0.00 ~ 10.00]											
Antenna:A Datarate: HT40-MCS7						Antenna:B Datarate: HT40-MCS7					
channel	3	7	11			channel	3	7	11		
SigLevel	-69	-69	-69			SigLevel	-69	-69	-69		
PER(%)	0.8	0.3	9.0			PER(%)	1.5	1.0	1.0		

5G
TX

标准: PktNum:1000; PER(%): [0.00 ~ 10.00]											
Antenna:A Datarate: OFDM-6						Antenna:B Datarate: OFDM-6					
channel	36	48	52	64	100	120	140	149	153	165	
Power	14.15	14.54	14.40	14.82	14.63	14.64	14.67	14.40	14.60	14.44	
Evm	33.99	32.36	33.63	33.58	33.79	32.75	32.72	32.40	31.65	31.81	
ppm	5.58	5.34	5.36	5.55	5.55	5.39	5.48	5.45	5.36	5.44	
Mask	0	0	0	0	0	0	0	0	0	0	
标准: PktNum:1000; PER(%): [0.00 ~ 10.00]											
Antenna:A Datarate: OFDM-54						Antenna:B Datarate: OFDM-54					
channel	36	48	52	64	100	120	140	149	153	165	
Power	14.27	14.70	14.51	14.90	14.71	14.77	14.78	14.48	14.62	14.52	
Evm	34.74	32.20	33.74	33.85	34.30	32.66	33.14	33.02	32.52	32.23	
ppm	5.31	5.33	5.30	5.45	5.42	5.34	5.47	5.53	5.35	5.40	
Mask	0	0	0	0	0	0	0	0	0	0	
标准: PktNum:1000; PER(%): [0.00 ~ 10.00]											
Antenna:A Datarate: HT20-MCS0						Antenna:B Datarate: HT20-MCS0					
channel	36	48	52	64	100	120	140	149	153	165	
Power	13.55	13.48	13.78	13.77	13.58	13.57	13.69	13.89	14.04	13.78	
Evm	35.10	34.78	34.95	34.81	34.20	32.79	33.06	32.58	32.67	32.67	
ppm	5.43	5.44	5.41	5.42	5.41	5.46	5.45	5.49	5.42	5.47	
Mask	0	0	0	0	0	0	0	0	0	0	
标准: PktNum:1000; PER(%): [0.00 ~ 10.00]											
Antenna:A Datarate: HT20-MCS3						Antenna:B Datarate: HT20-MCS3					
channel	36	48	52	64	100	120	140	149	153	165	
Power	13.64	13.60	13.95	13.89	13.72	13.75	13.82	14.04	14.20	13.93	
Evm	35.10	35.45	34.61	34.05	33.19	32.87	32.95	31.65	32.29	31.65	
ppm	5.56	5.43	5.44	5.43	5.43	5.45	5.45	5.45	5.46	5.44	
Mask	0	0	0	0	0	0	0	0	0	0	
标准: PktNum:1000; PER(%): [0.00 ~ 10.00]											
Antenna:A Datarate: HT20-MCS7						Antenna:B Datarate: HT20-MCS7					
channel	36	48	52	64	100	120	140	149	153	165	
Power	13.65	13.62	13.90	13.84	13.67	13.66	13.74	13.92	13.57	13.87	
Evm	35.48	35.16	34.71	35.03	34.28	32.57	32.22	32.96	31.31	32.91	
ppm	5.41	5.38	5.44	5.5	5.41	5.43	5.49	5.50	5.54	5.39	
Mask	0	0	0	0	0	0	0	0	0	0	



antenna channel 38 46 62 102 118 134 151 159																	antenna channel 38 46 62 102 118 134 151 159																
Power -12.29 -12.34 -12.36 -12.55 -12.35 -12.22 -12.32 -12.63																	Power -12.74 -12.65 -12.51 -12.68 -12.71 -12.52 -12.49 -12.54																
Evm -34.95 -34.50 -34.16 -33.44 -32.91 -32.56 -32.61 -32.22																	Evm -35.84 -33.91 -33.26 -33.48 -34.58 -34.94 -32.39 -33.24																
ppm -5.44 -5.40 -5.50 -5.42 -5.52 -5.49 -5.49 -5.47																	ppm -5.40 -5.39 -5.43 -5.42 -5.39 -5.46 -5.40 -5.38																
Mask 0 0 0 0 0 0 0 0																	Mask 0 0 0 0 0 0 0 0																
antenna channel 38 46 62 102 118 134 151 159																	antenna channel 38 46 62 102 118 134 151 159																
Power -12.48 -12.49 -12.56 -12.70 -12.58 -12.42 -12.56 -12.81																	Power -12.91 -12.82 -12.68 -12.86 -12.85 -12.67 -12.65 -12.71																
Evm -34.86 -34.50 -34.10 -33.77 -32.30 -32.38 -31.97 -31.98																	Evm -34.94 -34.35 -34.74 -35.76 -34.55 -35.01 -34.82 -34.22																
ppm -5.43 -5.36 -5.45 -5.53 -5.44 -5.42 -5.51 -5.45																	ppm -5.41 -5.38 -5.42 -5.45 -5.45 -5.41 -5.39 -5.41																
Mask 0 0 0 0 0 0 0 0																	Mask 0 0 0 0 0 0 0 0																
antenna channel 38 46 62 102 118 134 151 159																	antenna channel 38 46 62 102 118 134 151 159																
Power -13.90 -13.54 -13.57 -13.75 -13.55 -13.99 -13.54 -13.79																	Power -13.89 -13.79 -13.60 -13.74 -13.73 -13.58 -13.51 -13.57																
Evm -33.95 -34.41 -34.78 -33.66 -33.49 -32.96 -32.06 -32.57																	Evm -34.50 -32.94 -33.69 -35.42 -34.19 -34.99 -33.25 -34.05																
ppm -5.49 -5.49 -5.47 -5.52 -5.50 -5.44 -5.45 -5.53																	ppm -5.51 -5.51 -5.52 -5.52 -5.51 -5.55 -5.54 -5.58																
Mask 0 0 0 0 0 0 0 0																	Mask 0 0 0 0 0 0 0 0																
antenna channel 38 46 62 100 120 140 149 153 165																	antenna channel 38 46 62 100 120 140 149 153 165																
Power -13.64 -13.88 -13.83 -13.87 -13.89 -13.77 -13.84 -13.84																	Power -13.85 -13.82 -13.82 -13.92 -13.89 -13.82 -13.82 -13.99																
Evm -36.71 -36.96 -35.88 -35.20 -34.59 -34.57 -34.14 -33.93																	Evm -35.86 -33.76 -33.23 -37.02 -36.35 -34.57 -35.05 -34.79																
ppm -5.37 -5.40 -5.43 -5.37 -5.45 -5.35 -5.52 -5.40																	ppm -5.44 -5.37 -5.49 -5.33 -5.49 -5.48 -5.42 -5.36																
Mask 0 0 0 0 0 0 0 0																	Mask 0 0 0 0 0 0 0 0																
antenna channel 36 52 64 100 120 140 149 153 165																	antenna channel 36 52 64 100 120 140 149 153 165																
Power -13.71 -13.97 -13.95 -13.73 -13.76 -13.83 -14.10 -14.27																	Power -13.94 -13.88 -13.72 -13.97 -13.77 -14.40 -13.68 -13.91																
Evm -37.74 -37.16 -37.36 -37.39 -36.33 -35.73 -35.31 -34.60																	Evm -37.47 -34.01 -37.33 -37.66 -37.46 -36.32 -37.14 -36.31																
ppm -5.44 -5.45 -5.48 -5.44 -5.42 -5.47 -5.52 -5.50																	ppm -5.44 -5.40 -5.55 -5.47 -5.38 -5.48 -5.46 -5.41																
Mask 0 0 0 0 0 0 0 0																	Mask 0 0 0 0 0 0 0 0																
antenna channel 36 52 64 100 120 140 149 153 165																	antenna channel 36 52 64 100 120 140 149 153 165																
Power -13.63 -13.86 -13.84 -13.68 -13.67 -13.76 -13.97 -13.53																	Power -13.73 -13.69 -13.50 -13.80 -13.52 -13.78 -13.50 -13.68																
Evm -37.74 -38.18 -37.31 -36.94 -35.97 -35.71 -34.94 -34.84																	Evm -38.77 -34.71 -37.45 -39.60 -38.49 -38.12 -36.50 -36.68																
ppm -5.32 -5.42 -5.58 -5.46 -5.53 -5.46 -5.44 -5.53																	ppm -5.45 -5.55 -5.47 -5.47 -5.61 -5.43 -5.54 -5.44																
Mask 0 0 0 0 0 0 0 0																	Mask 0 0 0 0 0 0 0 0																
antenna channel 38 46 62 102 118 134 151 159																	antenna channel 38 46 62 102 118 134 151 159																
Power -12.31 -12.37 -12.40 -12.51 -12.41 -12.21 -12.38 -12.60																	Power -12.83 -12.74 -12.59 -12.68 -12.73 -12.53 -12.52 -12.51																
Evm -35.62 -34.46 -34.98 -34.46 -33.58 -33.81 -32.36 -32.70																	Evm -35.66 -33.10 -34.72 -34.79 -33.33 -35.55 -33.69 -34.23																
ppm -5.45 -5.45 -5.45 -5.45 -5.32 -5.51 -5.57 -5.42																	ppm -5.35 -5.34 -5.42 -5.56 -5.39 -5.43 -5.34 -5.33																
Mask 0 0 0 0 0 0 0 0																	Mask 0 0 0 0 0 0 0 0																
antenna channel 38 46 62 102 118 134 151 159																	antenna channel 38 46 62 102 118 134 151 159																
Power -12.93 -12.57 -12.62 -12.73 -12.62 -12.42 -12.57 -12.83																	Power -12.96 -12.92 -12.79 -12.85 -12.92 -12.71 -12.70 -12.70																
Evm -35.94 -35.98 -35.43 -35.25 -35.21 -35.02 -35.26 -34.64																	Evm -35.90 -35.18 -35.02 -35.02 -34.77 -34.93 -35.93 -37.22																
ppm -5.43 -5.29 -5.32 -5.57 -5.37 -5.35 -5.44 -5.65																	ppm -5.45 -5.35 -5.34 -5.45 -5.36 -5.30 -5.40 -5.55																
Mask 0 0 0 0 0 0 0 0																	Mask 0 0 0 0 0 0 0 0																
antenna channel 38 46 62 102 118 134 151 159																	antenna channel 38 46 62 102 118 134 151 159																
Power -12.54 -12.61 -12.60 -12.84 -12.67 -12.54 -12.61 -12.83																																	

5G
RX

Antenna: A channel 36 48 60 72 84 96 108 120 132 144 156 168 SigLevel -63 -63 -63 -63 -63 -63 -63 -63 -63 -63 -63 -63 PER(%) 0.60 0.50 0.6 0.60 0.60 0.70 0.2 4.10 0.40 1.90		Antenna: B channel 36 48 60 72 84 96 108 120 132 144 156 168 SigLevel -63 -63 -63 -63 -63 -63 -63 -63 -63 -63 -63 -63 PER(%) 0.90 0.50 0.5 0.30 0.20 0.4 1.3 0.60 1.3	
Antenna: A channel 36 48 60 72 84 96 108 120 132 144 156 168 SigLevel -86 -86 -86 -86 -86 -86 -86 -86 -86 -86 -86 -86 PER(%) 0.40 0.40 0 0 0.30 0 0 1.70 0.80		Antenna: B channel 36 48 60 72 84 96 108 120 132 144 156 168 SigLevel -86 -86 -86 -86 -86 -86 -86 -86 -86 -86 -86 -86 PER(%) 0.40 0.30 0 0 0 0 0 0.3 0.1	
Antenna: A channel 36 48 60 72 84 96 108 120 132 144 156 168 SigLevel -60 -60 -60 -60 -60 -60 -60 -60 -60 -60 -60 -60 PER(%) 1.50 1.50 1.0 0.1 0.5 0.3 0.1 0.7		Antenna: B channel 36 48 60 72 84 96 108 120 132 144 156 168 SigLevel -60 -60 -60 -60 -60 -60 -60 -60 -60 -60 -60 -60 PER(%) 2.50 2.10 2.00 0.70 0.9 0.40 1.0 0.5	
Antenna: A channel 42 58 74 90 106 122 138 154 170 186 202 218 SigLevel -63 -63 -63 -63 -63 -63 -63 -63 -63 -63 -63 -63 PER(%) 0.70 0.50 0 0.20 0.1 0.10		Antenna: B channel 42 58 74 90 106 122 138 154 170 186 202 218 SigLevel -63 -63 -63 -63 -63 -63 -63 -63 -63 -63 -63 -63 PER(%) 0.50 3.8 0.40 0.30 0.90 1.20	
Antenna: A channel 42 58 74 90 106 122 138 154 170 186 202 218 SigLevel -56 -56 -56 -56 -56 -56 -56 -56 -56 -56 -56 -56 PER(%) 0.40 0.10 0 0.10 0.20 2.40		Antenna: B channel 42 58 74 90 106 122 138 154 170 186 202 218 SigLevel -56 -56 -56 -56 -56 -56 -56 -56 -56 -56 -56 -56 PER(%) 1.10 0.60 0.20 0.20 0.90	

RF 最大最小值测试

测试前提

测试数据必须是测试至少 5 片样机后 RF 一致性 OK 的数据

结论

2G
TX

antenna	A		datarate	CCK-1	antenna	B		datarate	CCK-1
channel	1	7	13		channel	1	7	13	
Tpi	62	63	63		Tpi	62	63	63	
Power	23.50	24.08	23.96		Power	23.74	24.06	23.40	
Evm	-31.51	-27.43	-28.12		Evm	-30.96	-31.62	-28.59	
ppm	+9.70	+9.40	+9.44		ppm	+9.67	+9.67	+9.66	
Mask	4.67	2.08	0.25		Mask	3.67	2.42	2.75	
antenna	A		datarate	CCK-11	antenna	B		datarate	CCK-11
channel	1	7	13		channel	1	7	13	
Tpi	63	63	63		Tpi	63	63	63	
Power	24.01	24.26	24.10		Power	24.49	24.25	23.58	
Evm	-24.55	-23.44	-24.00		Evm	-25.12	-25.02	-25.26	
ppm	+9.81	+9.59	+9.69		ppm	+9.78	+9.77	+9.73	
Mask	1.83	0.08	0		Mask	0.83	0.25	1.25	
antenna	A		datarate	OFDM-54	antenna	B		datarate	OFDM-54
channel	1	7	13		channel	1	7	13	
Tpi	46	49	52		Tpi	51	52	53	
Power	17.17	18.39	19.33		Power	19.68	19.70	19.31	
Evm	-25.32	-25.83	-25.09		Evm	-25.25	-25.64	-25.94	
ppm	+10.26	+9.86	+9.64		ppm	+9.34	+9.15	+9.37	
Mask	0	0	0		Mask	0	0	0	
antenna	A		datarate	HT20-MCS0	antenna	B		datarate	HT20-MCS0
channel	1	7	13		channel	1	7	13	
Tpi	63	63	63		Tpi	63	63	63	
Power	22.97	23.20	23.12		Power	23.45	23.27	22.63	
Evm	-11.60	-12.63	-15.09		Evm	-16.62	-17.81	-18.69	
ppm	+8.74	+7.76	+8.78		ppm	+9.20	+9.39	+9.40	
Mask	0.08	0	0		Mask	0.08	0	0	



antenna	A		datarate	HT20-MCS3	antenna	B		datarate	HT20-MCS3
channel	1	7	13		channel	1	7	13	
Tpi	56	58	60		Tpi	63	63	63	
Power	20.98	21.86	22.35		Power	23.65	23.42	22.79	
Evm	-16.18	-16.95	-16.99		Evm	-16.38	-17.41	-18.35	
ppm	+9.91	+8.72	+8.28		ppm	+9.70	+10.04	+9.73	
Mask	0	0	0		Mask	0.25	0	0	
antenna	A		datarate	HT20-MCS7	antenna	B		datarate	HT20-MCS7
channel	1	7	13		channel	1	7	13	
Tpi	43	44	48		Tpi	48	49	51	
Power	15.73	16.16	17.67		Power	18.44	18.50	18.53	
Evm	-28.65	-28.44	-28.69		Evm	-28.15	-29.02	-28.36	
ppm	+9.99	+9.77	+9.67		ppm	+9.46	+9.77	+9.76	
Mask	0	0	0		Mask	0	0	0	
antenna	A		datarate	HT40-MCS0	antenna	B		datarate	HT40-MCS0
channel	3	7	11		channel	3	7	11	
Tpi	63	63	63		Tpi	63	63	63	
Power	23.12	23.36	23.34		Power	23.55	23.37	23.02	
Evm	-15.09	-15.14	-16.35		Evm	-16.63	-17.29	-17.88	
ppm	+8.78	+11.06	+11.02		ppm	+10.06	+9.90	+9.86	
Mask	0	2.33	0.50		Mask	3.67	1.33	0.42	
antenna	A		datarate	HT40-MCS3	antenna	B		datarate	HT40-MCS3
channel	3	7	11		channel	3	7	11	
Tpi	62	60	62		Tpi	62	63	63	
Power	22.97	22.80	23.36		Power	23.56	23.64	23.25	
Evm	-14.59	-16.08	-16.24		Evm	-16.73	-16.90	-17.42	
ppm	+10.92	+10.65	+10.97		ppm	+10.15	+10.18	+9.90	
Mask	4.33	0.33	0.50		Mask	2.92	2.17	1.25	
antenna	A		datarate	HT40-MCS7	antenna	B		datarate	HT40-MCS7
channel	3	7	11		channel	3	7	11	
Tpi	45	47	47		Tpi	47	48	49	
Power	17.05	17.96	17.81		Power	18.26	18.47	18.35	
Evm	-28.34	-28.10	-29.07		Evm	-28.37	-28.33	-28.69	
ppm	+9.45	+9.61	+9.70		ppm	+9.24	+9.36	+9.49	
Mask	0	0	0		Mask	0	0	0	



2G
RX

A	1	3	7	11	13
CCK-1	-5/-96		-5/-96		-5/-97
CCK-11	-5/-87		-5/-88		-5/-88
OFDM-54	-5/-74		-5/-74		-5/-74
HT20-MCS0	-5/-89		-5/-90		-5/-91
HT20-MCS7	-5/-71		-5/-72		-5/-72
HT40-MCS0		-5/-89	-5/-89	-5/-89	
HT40-MCS7		-5/-70	-5/-70	-5/-71	
B	1	3	7	11	13
CCK-1	-5/-96		-5/-96		-5/-96
CCK-11	-5/-86		-5/-87		-5/-87
OFDM-54	-5/-74		-5/-74		-5/-74
HT20-MCS0	-5/-89		-5/-90		-5/-91
HT20-MCS7	-5/-71		-5/-72		-5/-72
HT40-MCS0		-5/-89	-5/-89	-5/-89	
HT40-MCS7		-5/-70	-5/-70	-5/-70	

5G
TX[illegible]

深圳市磊科实业有限公司

	antenna A channel 36 52 64 100 120 140 149 153 165 Tpi 58 60 61 57 57 58 57 Power 51.15 51.14 52.35 52.13 51.37 51.30 51.15 51.00 50.82 Evm 51.59 51.09 51.88 51.04 51.64 51.55 51.30 51.80 51.81 ppm 4.07 4.00 3.91 3.96 3.98 3.93 3.81 3.85 Mask 5.92 5.75 4.67 5.08 4.42 5.25 4.38 5.25 5.42																			
	antenna B channel 36 52 64 100 120 140 149 153 165 Tpi 58 61 61 59 59 60 62 61 Power 51.43 51.12 51.69 51.79 51.48 51.66 51.36 51.17 51.41 Evm 51.86 51.12 51.62 51.04 51.29 51.73 51.57 51.16 51.03 ppm 4.66 4.73 4.85 4.96 4.89 4.82 4.75 4.82 4.72 Mask 4.75 4.92 4.08 4.30 5.25 4.58 4.08 4.33 4.00																			
	antenna A channel 36 52 64 100 120 140 149 153 165 Tpi 54 55 56 54 55 55 59 57 Power 51.94 51.83 51.12 51.44 50.55 50.42 50.27 50.12 51.88 Evm 51.45 51.27 51.28 51.05 50.22 51.34 51.03 51.41 51.30 ppm 5.84 5.79 5.16 5.82 5.65 5.87 5.65 5.63 5.7 Mask 6.25 6.17 6.25 5.33 5.08 5.58 5.42 6.6 6.33																			
	antenna B channel 36 52 64 100 120 140 149 153 165 Tpi 45 45 45 45 45 45 45 45 45 Power 51.22 51.32 51.47 51.35 51.97 51.02 51.33 51.56 51.49 Evm 51.94 51.50 51.89 51.97 51.48 51.21 51.41 51.31 51.88 ppm 5.74 5.64 5.84 5.77 5.71 5.78 5.69 5.80 5.68 Mask 0 0 0 0 0 0 0 0 0																			
	antenna A channel 38 46 62 102 118 134 151 155 Tpi 53 53 53 53 53 53 53 53 53 Power 51.44 51.44 50.69 51.01 51.87 50.35 51.70 51.80 Evm 51.23 51.59 51.08 51.36 51.73 51.69 51.74 51.69 ppm 3.60 3.76 3.76 3.74 3.79 3.86 3.85 3.86 Mask 2.67 4.67 4.67 5.83 4.75 5.50 4.82 4.42																			
	antenna B channel 38 46 62 102 118 134 151 155 Tpi 53 53 53 53 53 53 53 53 53 Power 51.43 51.43 51.88 51.82 50.49 50.11 50.97 51.79 Evm 51.01 51.30 51.36 51.18 51.97 51.68 51.78 51.88 ppm 4.78 4.74 4.81 4.78 4.71 4.73 4.76 4.81 Mask 4.42 4.25 5.08 5.83 4.42 4.83 5.83 5.25																			
	antenna A channel 38 46 62 102 118 134 151 155 Tpi 45 45 45 45 45 45 45 45 45 Power 51.09 51.13 51.76 51.80 51.73 51.57 51.57 51.70 Evm 51.30 51.34 51.36 51.78 51.32 51.90 51.57 51.46 ppm 3.68 3.76 3.76 3.69 3.71 3.73 3.75 3.79 Mask 0 0 0 0 0 0 0 0																			
	antenna B channel 38 46 62 102 118 134 151 155 Tpi 45 45 45 45 45 45 45 45 45 Power 51.09 51.13 51.76 51.80 51.73 51.57 51.57 51.70 Evm 51.30 51.34 51.36 51.78 51.32 51.90 51.57 51.46 ppm 3.68 3.76 3.76 3.69 3.71 3.73 3.75 3.79 Mask 0 0 0 0 0 0 0 0																			
	antenna A channel 42 58 106 122 138 155 Tpi 63 63 63 63 63 63 63 63 63 Power 51.31 51.99 51.74 51.76 51.70 51.16 51.16 Evm 51.82 51.13 51.60 51.02 51.15 51.83 ppm 5.76 5.87 4.14 5.99 4.13 4.17 Mask 6.92 6.7 6.92 6.92 6.83 6.42																			
	antenna B channel 42 58 106 122 138 155 Tpi 54 53 53 53 53 53 53 53 53 Power 51.91 51.68 51.68 51.65 51.58 51.56 Evm 51.18 51.41 51.41 51.42 51.06 51.35 ppm 4.64 4.70 4.70 4.75 4.79 4.68 Mask 0 0 0 0 0 0																			
	antenna A channel 42 58 106 122 138 155 Tpi 54 53 53 53 53 53 53 53 53 Power 51.91 51.68 51.68 51.65 51.58 51.56 Evm 51.18 51.41 51.41 51.42 51.06 51.35 ppm 4.64 4.70 4.70 4.75 4.79 4.68 Mask 0 0 0 0 0 0																			
	antenna B channel 42 58 106 122 138 155 Tpi 54 53 53 53 53 53 53 53 53 Power 51.91 51.68 51.68 51.65 51.58 51.56 Evm 51.18 51.41 51.41 51.42 51.06 51.35 ppm 4.64 4.70 4.70 4.75 4.79 4.68 Mask 0 0 0 0 0 0																			

5G RX	antenna A channel 36 52 64 100 120 140 149 153 165 Tpi 58 60 61 57 57 58 57 Power 51.15 51.14 52.35 52.13 51.37 51.30 51.15 51.00 50.82 Evm 51.59 51.09 51.88 51.04 51.64 51.55 51.30 51.80 51.81 ppm 4.07 4.00 3.91 3.96 3.98 3.93 3.81 3.85 Mask 5.92 5.75 4.67 5.08 4.42 5.25 4.38 5.25 5.42																			
	antenna B channel 36 52 64 100 120 140 149 153 165 Tpi 58 61 61 59 59 60 62 61 Power 51.43 51.12 51.69 51.79 51.48 51.66 51.36 51.17 51.41 Evm 51.86 51.12 51.62 51.04 51.29 51.73 51.57 51.16 51.03 ppm 4.66 4.73 4.85 4.96 4.89 4.82 4.75 4.82 4.72 Mask 4.75 4.92 4.08 4.30 5.25 4.58 4.08 4.33 4.00																			
	antenna A channel 36 52 64 100 120 140 149 153 165 Tpi 54 55 56 54 55 55 59 57 Power 51.94 51.83 51.12 51.44 50.55 50.42 50.27 50.12 51.88 Evm 51.45 51.27 51.28 51.05 50.22 51.34 51.03 51.41 51.30 ppm 5.84 5.79 5.16 5.82 5.65 5.87 5.65 5.63 5.7 Mask 6.25 6.17 6.25 5.33 5.08 5.58 5.42 6.6 6.33																			
	antenna B channel 36 52 64 100 120 140 149 153 165 Tpi 45 45 45 45 45 45 45 45 45 Power 51.22 51.32 51.47 51.35 51.97 51.02 51.33 51.56 51.49 Evm 51.94 51.50 51.89 51.97 51.48 51.21 51.41 51.31 51.88 ppm 5.74 5.64 5.84 5.77 5.71 5.78 5.69 5.80 5.68 Mask 0 0 0 0 0 0 0 0 0																			
	antenna A channel 38 46 62 102 118 134 151 155 Tpi 53 53 53 53 53 53 53 53 53 Power 51.44 51.44 50.69 51.01 51.87 50.35 51.70 51.80 Evm 51.23 51.59 51.08 51.36 51.73 51.69 51.74 51.69 ppm 3.60 3.76 3.76 3.74 3.79 3.86 3.85 3.86 Mask 2.67 4.67 4.67 5.83 4.75 5.50 4.82 4.42																			
	antenna B channel 38 46 62 102 118 134 151 155 Tpi 53 53 53 53 53 53 53 53 53 Power 51.43 51.43 51.88 51.82 50.49 50.11 50.97 51.79 Evm 51.01 51.30 51.36 51.18 51.97 51.68 51.78 51.88 ppm 4.78 4.74 4.81 4.78 4.71 4.73 4.76 4.81 Mask 4.42 4.25 5.08 5.83 4.42 4.83 5.83 5.25																			
	antenna A channel 38 46 62 102 118 134 151 155 Tpi 45 45 45 45 45 45 45 45 45 Power 51.09 51.13 51.76 51.80 51.73 51.57 51.57 51.70 Evm 51.30 51.34 51.36 51.78 51.32 51.90 51.57 51.46 ppm 3.68 3.76 3.76 3.69 3.71 3.73 3.75 3.79 Mask 0 0 0 0 0 0 0 0																			
	antenna B channel 38 46 62 102 118 134 151 155 Tpi 45 45 45 45 45 45 45 45 45 Power 51.09 51.13 51.76 51.80 51.73 51.57 51.57 51.70 Evm 51.30 51.34 51.36 51.78 51.32 51.90 51.57 51.46 ppm 3.68 3.76 3.76 3.69 3.71 3.73 3.75 3.79 Mask 0 0 0 0 0 0 0 0																			
	antenna A channel 42 58 106 122 138 155 Tpi 63 63 63 63 63 63 63 63 63 Power 51.31 51.99 51.74 51.76 51.70 51.16 51.16 Evm 51.82 51.13 51.60 51.02 51.15 51.83 ppm 5.76 5.87 4.14 5.99 4.13 4.17 Mask 6.92 6.7 6.92 6.92 6.83 6.42																			
	antenna B channel 42 58 106 122 138 155 Tpi 54 53 53 53 53 53 53 53 53 Power 51.91 51.68 51.68 51.65 51.58 51.56 Evm 51.18 51.41 51.41 51.42 51.06 51.35 ppm 4.64 4.70 4.70 4.75 4.79 4.68 Mask 0 0 0 0 0 0																			
	antenna A channel 42 58 106 122 138 155 Tpi 54 53 53 53 53 53 53 53 53 Power 51.91 51.68 51.68 51.65 51.58 51.56 Evm 51.18 51.41 51.41 51.42 51.06 51.35 ppm 4.64 4.70 4.70 4.75 4.79 4.68 Mask 0 0 0 0 0 0																			
	antenna B channel 42 58 106 122 138 155 Tpi 54 53 53 53 53 53 53 53 53 Power 51.91 51.68 51.68 51.65 51.58 51.56 Evm 51.18 51.41 51.41 51.42 51.06 51.35 ppm 4.64 4.70 4.70 4.75 4.79 4.68 Mask 0 0 0 0 0 0																			

5G RX	antenna A channel 36 52 64 100 120 140 149 153 165 Tpi 58 60 61 57 57 58 57 Power 51.15 51.14 52.35 52.13 51.37 51.30 51.15 51.00 50.82 Evm 51.59 51.09 51.88 51.04 51.64 51.55 51.30 51.80 51.81 ppm 4.07 4.00 3.91 3.96 3.98 3.93 3.81 3.85 Mask 5.92 5.75 4.67 5.08 4.42 5.25 4.38 5.25 5.42																			
	antenna B channel 36 52 64 100 120 140 149 153 165 Tpi 58 61 61 59 59 60 62 61 Power 51.43 51.12 51.69 51.79 51.48 51.66 51.36 51.17 51.41 Evm 51.86 51.12 51.62 51.04 51.29 51.73 51.57 51.16 51.03 ppm 4.66 4.73 4.85 4.96 4.89 4.82 4.75 4.82 4.72 Mask 4.75 4.92 4.08 4.30 5.25 4.58 4.08 4.33 4.00																			
	antenna A channel 36 52 64 100 120 140 149 153 165 Tpi 54 55 56 54 55 55 59 57 Power 51.94 51.83 51.12 51.44 50.55 50.42 50.27 50.12 51.88 Evm 51.45 51.27 51.28 51.05 50.22 51.34 51.03 51.41 51.30 ppm 5.84 5.79 5.16 5.82 5.65 5.87 5.65 5.63 5.7 Mask 6.25 6.17 6.25 5.33 5.08 5.58 5.42 6.6 6.33																			
	antenna B channel 36 52 64 100 12																			

按键未做功能。

3.2、串口

串口速率：38400 测试 OK。

4、性能测试

4.1.1、无线吞吐量测试（网络接口的连接速度为 1G）

2.4G

#11	40M (300Mbps)			20M (144.5Mbps)		
	1	7	13	1	7	13
TX	198.885	195.541	197.379	103.177	104.241	107.852
RX	213.825	213.611	213.682	107.828	110.185	111.935
TX+RX	212.952	215.868	214.347	105.469	101.829	111.213
#17	40M (300Mbps)			20M (144.5Mbps)		
	1	7	13	1	7	13
TX	201.465	199.131	200.291	100.852	106.371	105.251
RX	213.203	213.151	213.203	105.366	111.138	109.839
TX+RX	215.476	214.305	215.335	104.169	108.802	101.506
#12	40M (300Mbps)			20M (144.5Mbps)		
	1	7	13	1	7	13
TX	195.149	192.932	195.684	103.621	106.002	102.641
RX	211.496	211.784	212.584	108.692	111.572	111.841
TX+RX	213.027	212.104	213.168	108.152	104.177	110.669

5G



#17	80M(867Mbps)			40M(400Mbps)			20M(173.0Mbps)		
	36	149	161	36	149	161	36	149	161
TX	577.661	554.704	556.536	285.198	281.382	265.401	133.328	138.045	133.889
RX	526.121	513.402	484.555	307.051	293.382	270.188	143.581	143.318	132.318
TX+RX	534.036	514.591	512.319	299.853	292.994	286.738	142.317	142.753	134.163
#9	80M(867Mbps)			40M(400Mbps)			20M(173.0Mbps)		
	36	149	161	36	149	161	36	149	161
TX	554.356	548.147	546.501	284.391	284.391	284.017	135.448	137.372	136.131
RX	527.437	538.552	548.433	304.378	302.778	304.178	143.569	144.023	140.591
TX+RX	534.472	531.632	540.152	298.938	298.408	298.338	138.989	141.648	140.441
#12	80M(867Mbps)			40M(400Mbps)			20M(173.0Mbps)		
	36	149	161	36	149	161	36	149	161
TX	575.775	562.426	567.981	285.789	276.253	267.284	128.161	133.745	132.154
RX	544.132	526.023	521.461	308.541	307.237	282.657	143.981	143.361	135.282
TX+RX	550.467	506.221	517.086	296.902	288.691	274.604	140.691	131.217	133.265

4.2.2 两屏蔽箱间各衰减下的吞吐量测试

2.4G:

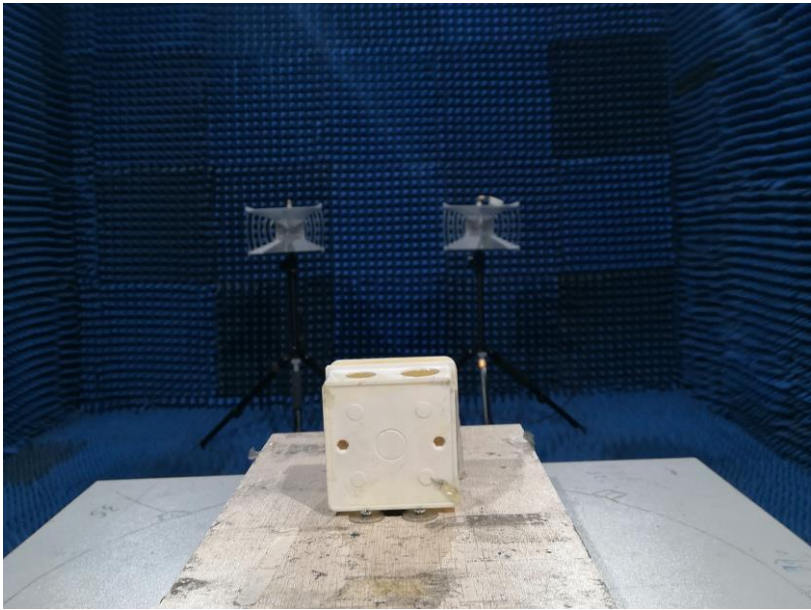
40M 自动信道 直连									
#17	30db	40db	50db	60db	70db	80db	90db	95db	95db
TX	201.506	202.522	193.242	189.472	149.832	94.373	18.821	ping	ping断
RX	212.124	212352	210.654	211.612	184.384	92.435	38.525	21.677	ping断
TX+RX	216.413	212.227	206.416	206.262	169.008	101.056	24.595	6.183	ping断
40M 自动信道 自带天线									
#17	0db	10db	20db	30db	40db	50db	60db	65db	
TX	199.572	202.207	198.985	200.091	164.125	104.134	47.562	ping	
RX	211.904	212.681	211.154	212.134	109.197	43.893	ping	ping	
TX+RX	214.114	214.361	215.119	212.551	141.779	63.219	12.079	ping	

5G:



80M 自动信道 直连								
#17	30db	40db	50db	60db	70db	80db	90db	95db
TX	580.001	581.653	570.125	570.333	437.771	208.641	110.772	35.713
RX	552.077	547.571	543.552	548.077	504.396	307.048	82.124	27.637
TX+RX	549.191	551.249	545.052	548.622	491.787	282.504	89.353	29.895
	100db	103db	105db					
TX	9.006	0.913	ping断					
RX	5.839	2.055	ping断					
TX+RX	6.377	1.422	ping断					
80M 自动信道 自带天线								
#17	0db	10db	20db	30db	40db	50db	60db	65db
TX	579.731	574.386	522.403	408.395	309.379	152.319	8.54	ping
RX	550.151	536.535	539.265	519.285	421.911	254.831	57.181	ping
TX+RX	550.339	542.943	509.776	429.802	360.842	201.914	32.471	ping

4.3、天线场型测试

环境		测试结论
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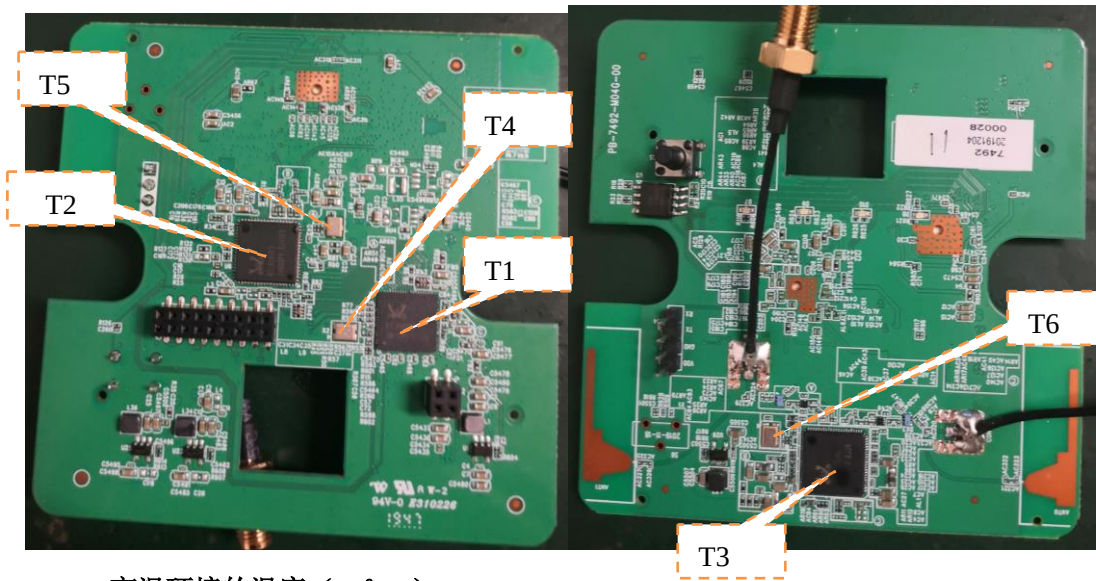
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2G	天线型号	PB-7492-2.4G						
	衰减	40	40	40	50	50	50	
	角度\通道	1	7	13	1	7	13	
	30	52.00892	55.26474	60.98909	20.50319	14.69862	17.44998	
	60	30.03727	40.25365	45.03045	14.02963	17.39846	15.88501	
	90	32.51507	39.18709	20.9814	15.6012	14.57486	13.64988	
	120	25.89752	38.69294	39.51303	15.04416	16.90845	16.89812	
	150	45.41744	43.50809	44.69948	12.21893	15.24368	14.0091	
	180	45.62908	53.57193	45.45263	0.317933	1.26348	0.991142	
	210	17.37136	25.2412	35.8764	0.091833	0	0	
	240	45.01167	57.07064	48.43119	0.52911	7.513047	2.248887	
	270	46.28317	61.42675	38.426	8.075124	13.05481	4.933576	
	300	39.10329	28.61053	18.98824	8.803532	17.60637	8.596803	
	330	27.5648	27.72916	22.12737	14.67311	17.77338	11.56106	
	360	54.69778	59.27187	66.91323	20.18576	15.23814	15.82153	
5G	天线型号	PB-7492-5G						
	衰减	30	30	30	30	40	40	40
	角度\通道	36	64	149	161	36	64	149
	30	89.62638	91.53318	87.22915	80.66203	18.35291	18.49376	15.85427
	60	113.0143	136.9086	122.4599	101.1193	27.4156	28.56426	24.33835
	90	73.7837	32.39335	55.62586	55.66046	26.25741	28.40886	25.12333
	120	173.3819	64.01011	115.286	119.9541	38.04215	45.14455	24.66913
	150	106.0779	110.8251	102.9576	123.3772	7.401381	25.21265	18.81388
	180	164.677	144.5758	127.3421	143.0346	1.571977	20.57339	2.668433
	210	115.6632	148.5646	144.5598	145.5535	12.45849	23.54409	16.71253
	240	136.7154	139.539	148.0648	145.1182	33.32502	33.32401	36.40735
	270	57.19967	23.80715	56.78119	53.70316	28.2583	13.7155	30.24387
	300	102.5896	50.53434	128.2791	103.4667	30.63929	30.11801	27.25863
	330	112.5667	102.0002	83.70698	95.44452	17.38516	24.14586	16.74628
	360	168.5192	162.56	139.7387	150.0953	22.56656	30.30339	25.11592

5、环境测试

5.1、DUT 高低温功能及温度





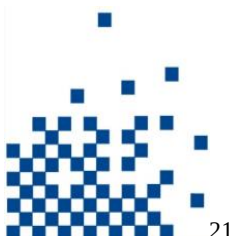
5.1.1、高温环境的温度 (50° C)

同时跑 LAN 口到 5G、2.4G 的 TX+RX 各 10 条流，5G 10 条流跑 280M 左右、2.4G 10 条流跑 200 M 左右，测试数据如下：

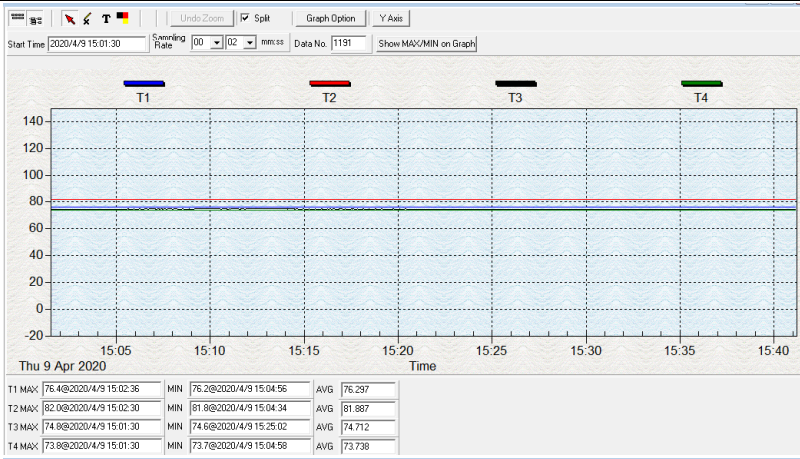
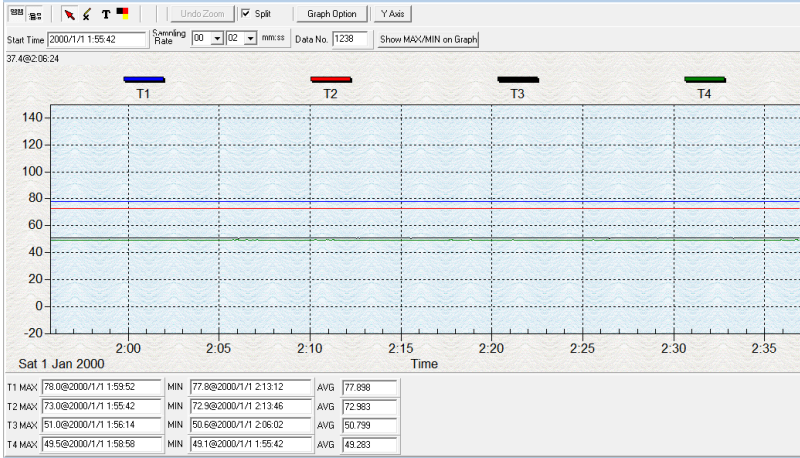
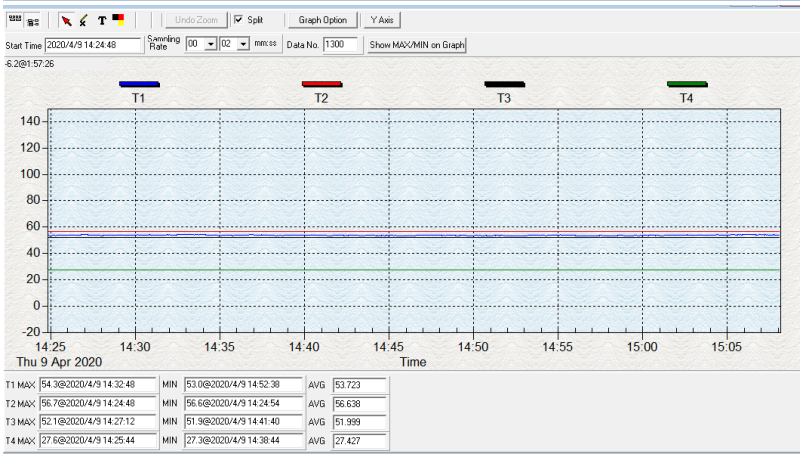
测试点编号	测试点	最高温度(℃)	温度曲线	外观有 无异常
T1	RTL8363N B-VB-CG (U4)	107.9		无
T2	RTL8197F S-VE5-CG (U6)	116.6		
T3	RTL8812B RH(U5)	114.4		
T4	晶振 (U4)	105.1		
T1	晶振 (U6)	112.3		
T2	晶振 (U5)	109.7		
T3	上盖 (U4)	78.2		
T4	上盖 (U6)	75.1		
T1	上盖 (U5)	85.1		
T2	下盖 (中间)	84.5		
T3	下盖 (电感)	79.5		
T4	环境	51.2		

5.1.2、常温环境的温度 (25° C)

移动标准：在常温下跑 LAN 口到 2.4G 的 TX+RX 各 10 条流， 2.4G 10 条流跑 200 M 左右，测试数据如

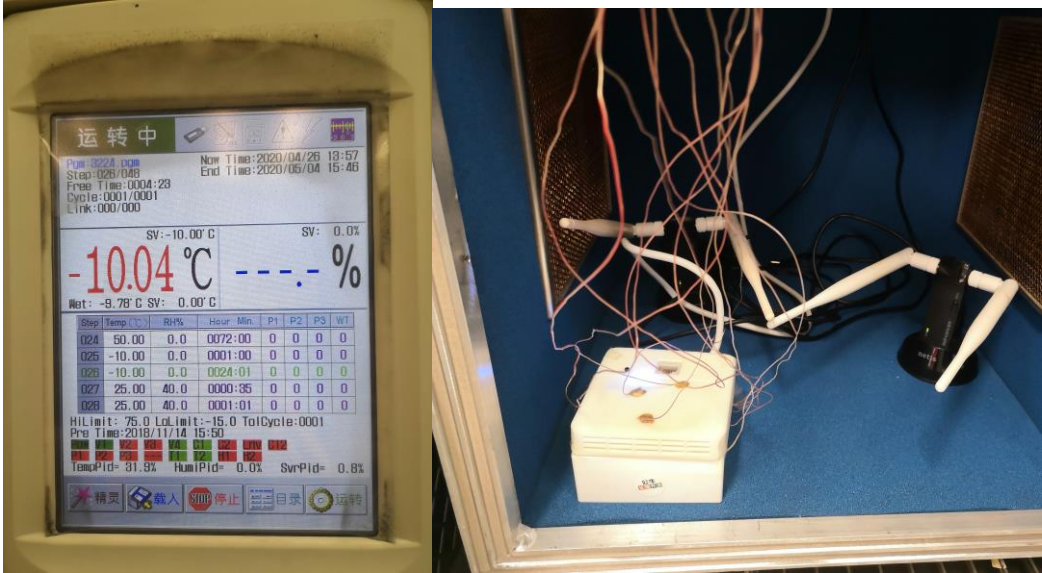


下:

测试点 编号	测试点	最高温 度(℃)	温度曲线	外观有 无异常
T1	RTL8363N B-VB-CG (U4)	76.4		无
T2	RTL8197F S-VE5-CG (U6)	82.0		
T3	RTL8812B RH(U5)	74.8		
T4	晶振 (U4)	73.8		
T1	晶振 (U6)	78.0		
T2	晶振 (U5)	73.0		
T3	上盖 (U4)	51.0		
T4	上盖 (U6)	49.5		
T1	上盖 (U5)	54.3		
T2	下盖 (中 间)	56.7		
T3	下盖 (电 感)	52.1		
T4	环境	27.6		

5.2 低温启动测试

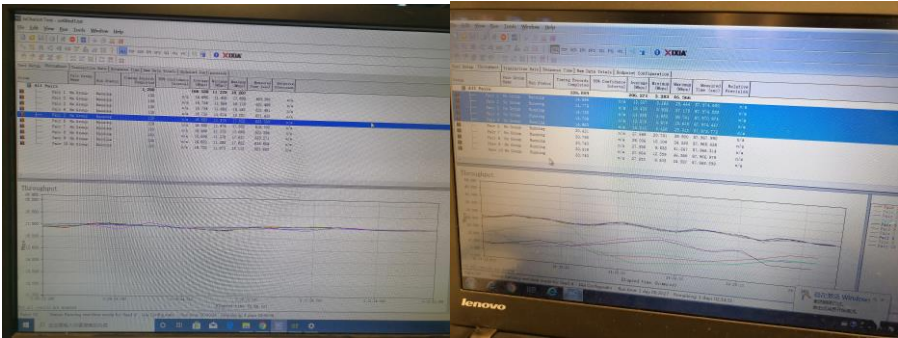


测试项目	低温启动测试
测试方法	将高低温箱温度设置下降至-10℃，被测设备不上电放入箱内持续 30 分钟左右，然后上电启动设备，设备完全启动后再次断电，循环重复 10 次以上下电启动，观察设备是否有不启动、死机等现象。
测试环境	
测试结论	机壳未出现异常，板子启动正常、性能正常

5.3 高温烧机

测试项目	高温烧机测试
测试方法	在高温 50℃环境下连续满载工作三天，3 台 PC 通过 chariot 软件跑 2G/5G High throughout，查看设备长时间高温工作是否出现断线或者死机等现象。

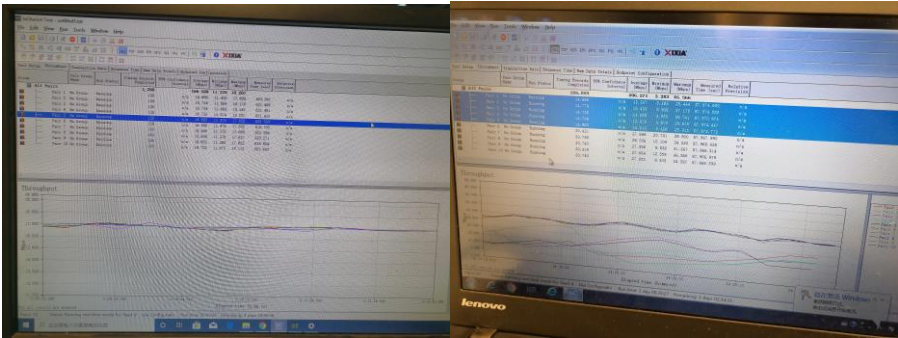
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测试环境	
测试数据	
测试结论	机壳未出现异常，板子启动正常、性能正常

5.4 低温运行测试

测试项目	低温运行测试
测试方法	在低温-10℃环境下连续满载工作 24H，3 台 PC 通过 chariot 软件跑 2G/5G High throughput，查看设备长时间低温工作是否出现断线或者死机等现象。

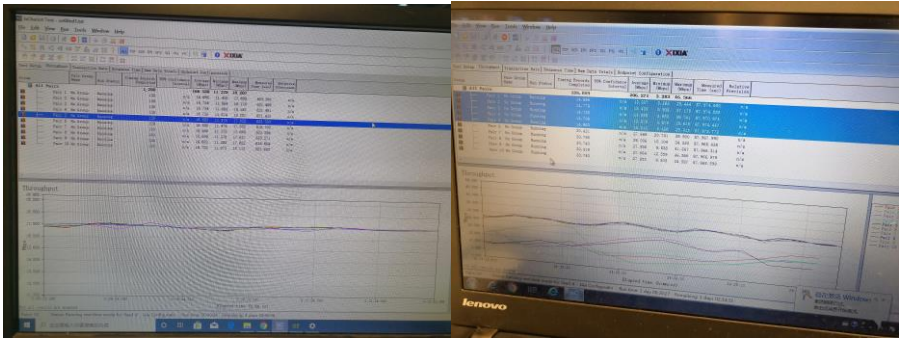
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测试环境	
测试数据	
测试结论	机壳未出现异常，板子启动正常、性能正常

5.5 温度循环及交变湿热测试

测试项目	温度循环及交变湿热测试
测试方法	25° ±3（持续 1 小时）→→ 50° （持续 24 小时）→→ -10° （持续 24 小时）→→ 25° 40±3%湿度（持续 1 小时）→→ 30° 93±3%湿度（持续 12 小时）→→ 40° 85±3%湿度（持续 12 小时）→→ 25° 95±3%的湿度（持续 12 小时）→→ 50° 95±3%湿度（持续 12 小时）→→ 25° （持续 1 小时）

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测试环境	
测试数据	
测试结论	机壳未出现异常，板子启动正常、性能正常

5.6 高温贮存测试

测试项目	高温贮存测试
测试方法	将设备空载下电放置在温度为 70℃ 的温箱内贮存 24H，试验结束后取出设备放置在常温下，待设备恢复至常温后观察设备结构外观是否正常，测试设备基本功能是否正常，满载性能是否正

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测试环境	
测试数据	
测试结论	机壳未出现异常，板子启动正常、性能正常

5.7 低温贮存测试

测试项目	低温贮存测试
测试方法	将设备空载下电放置在温度为-40℃的温箱内贮存 24H，试验结束后取出设备放置在常温下，待设备恢复至常温后观察设备结构外观是否正常，测试设备基本功能是否正常，满载性能是否正

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测试环境	
测试数据	
测试结论	机壳未出现异常，板子启动正常、性能正常

5.8、上下电测试

测试记录	WiFiTest - Performance net-core 磊科 SN 0145 0810740004e3 0.0.0.0 21857ms. Ping 通 Ping: 192.168.8.1...OK Telnet>echo 0 > /proc/gpio Telnet>echo 0 > /proc/gpio # Sleep (25000). Telnet>echo 1 > /proc/gpio Telnet>echo 1 > /proc/gpio # Sleep (29700). Telnet>echo 0 > /proc/gpio Telnet>echo 0 > /proc/gpio # Sleep (23500). ***** 第五次循环，第398测试。 ***** Telnet>echo 1 > /proc/gpio Telnet>echo 1 > /proc/gpio # Sleep (25000). 21857ms. Ping 通 Ping: 192.168.8.1...OK Telnet>echo 0 > /proc/gpio Telnet>echo 0 > /proc/gpio # Sleep (24000). Telnet>echo 1 > /proc/gpio Telnet>echo 1 > /proc/gpio # Sleep (39800). Telnet>echo 0 > /proc/gpio Telnet>echo 0 > /proc/gpio # Sleep (24000). ***** 第五次循环，第399测试。 ***** Telnet>echo 1 > /proc/gpio Telnet>echo 1 > /proc/gpio # Sleep (2000). 21856ms. Ping 通 Ping: 192.168.8.1...OK Telnet>echo 0 > /proc/gpio Telnet>echo 0 > /proc/gpio # Sleep (24500). Telnet>echo 1 > /proc/gpio Telnet>echo 1 > /proc/gpio # Sleep (39800). Telnet>echo 0 > /proc/gpio Telnet>echo 0 > /proc/gpio # Sleep (24500). ***** 第五次循环，第400测试。 ***** Telnet>echo 1 > /proc/gpio Telnet>echo 1 > /proc/gpio # Sleep (2000). 21856ms. Ping 通 Ping: 192.168.8.1...OK Telnet>echo 0 > /proc/gpio Telnet>echo 0 > /proc/gpio # Sleep (25000). Telnet>echo 1 > /proc/gpio Telnet>echo 1 > /proc/gpio # Sleep (40000). Telnet>echo 0 > /proc/gpio Telnet>echo 0 > /proc/gpio # Sleep (25000). ***** TestTime: 59:58:14.281 sec Result: Pass Statistics: Time: 59:58:08, Count: 3 (pass: 1, fail: 2, stop: 0) >>> Pass, Spend 215774.4 seconds, RealSpend: 59:58:23 Auto-SN IF Connect IF AutoRun PASS Start Windows - 虚拟内存大小过低 您的系统虚拟内存太低。Windows 会增加虚拟机内存页面文件的大小。在这个过程中，一些应用程序的内存请求会被拒绝。有关详细信息，请参阅“帮助”。 开始 命令提示符 - pin... 上下电测试 WiFiTest - Perf... 9:23
结果	PASS
备注	/

